Sora Kim

Regulatory HCI · STS · Computational Methods · Accountability in Regulated Systems

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RESEARCH FOCUS

I study how accountability converges prematurely in regulated organizations — how responsibility attribution settles on fast, documentable explanations before alternative causal accounts have been adequately examined. Grounded in four years of pharmaceutical GMP field experience, I developed this first as a structural theoretical claim about investigations (Responsibility Convergence, RC), and then examined how AI-mediated systems can intensify this tendency (Premature Accountability Convergence, PAC) by compressing interpretive space earlier in the sensemaking sequence. To move from theory to intervention, I operationalize convergence as a process-quality signal — focusing on detecting when reasoning closes, not merely that it has closed — and investigate whether interface and AI scaffolding can help preserve comparison at the point where attribution would otherwise prematurely settle. My goal is to design accountability-aware AI systems for regulated environments: systems that preserve the conditions under which meaningful human judgment remains possible, without altering regulatory constraints.

Interests: Regulatory HCI · STS · computational methods · accountability · regulated organizations · AI-mediated reasoning · document infrastructure · judgment-preserving design

RESEARCH PROGRAM

Theoretical contributions AAC – CF – RC – PAC lifecycle: a four-stage account of how responsibility is prospectively aligned before action, procedurally embedded during execution, retrospectively stabilized under closure pressure, and further shaped through AI-mediated representation.

Computational validation FDA Warning Letter corpus (N=50, 160 violations, 2016–2025): preliminary models suggest strong separability in PAC detection (AUC $>$ 0.90); accountability graphs exhibit centralized structures (Firm in-degree centrality = 0.444); responsibility attribution models achieve moderate performance (macro F1 $\approx$ 0.72). Across cases, responsibility attribution frequently converges toward firm-level explanations early in the document structure, suggesting structural pre-convergence prior to extended investigative reasoning.

Empirical findings across four HCI studies (N=20 GMP-experienced participants each):

- *Accessibility layer*: keyword-based retrieval substantially reduced exploratory SOP search time (from $\approx$6 min to $\approx$1 min), suggesting that access constraints limit the available behavioral repertoire.
- *Structural layer*: loopback behavior was consistently observed across sessions (mean $\approx$5 per session); less experienced practitioners exhibited more frequent loopbacks (7–10 vs. 2–3), suggesting a structural rather than purely cognitive source of navigation difficulty.
- *Coordination layer*: participants reported improved visibility of coordination burden and anticipated reduced accountability asymmetry; findings suggest a distinction between immediate usability and downstream interpretability.
- *Reasoning layer*: many participants initially produced single-cause explanations under closure pressure; AI scaffolding was associated with broader causal exploration and increased consideration of systemic factors. Usability and perceived usefulness were consistently high across participants.

SELECTED PUBLICATIONS

- STS **Responsibility Convergence in Organizational Investigations**
  SSRN Preprint, 2026
  Introduces responsibility convergence as a structural property of organizational investigations, arguing that under closure pressure, attribution tends to narrow toward fast, documentable explanations before sufficient epistemic exploration.
- STS **Premature Accountability Convergence in Regulated Organizations**
  SSRN Preprint, 2026
  Introduces PAC as a distinct failure mode, suggesting that AI representations can accelerate pre-existing convergence tendencies by compressing interpretive space before alternative accounts remain viable.
- Comp **Responsibility as a Multi-Layer Signal: A Unified Computational Framework for Accountability Convergence in Regulatory Investigations**
  arXiv Preprint, 2026
  Proposes a unified computational framework integrating PAC detection, accountability graph extraction, and responsibility attribution modeling. Applied to FDA Warning Letters, results suggest that accountability structure is partially encoded in document architecture prior to extended reasoning.
- HCI **Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations**
  SSRN Preprint, 2026
  Reframes premature convergence as an interaction problem and investigates whether interface scaffolding can preserve comparison across causal alternatives during investigation. Findings suggest increased exploration of systemic explanations under scaffolded conditions.

WORKING PAPERS

- STS **Anticipated Accountability Convergence: Structuring Responsibility Before Action in Regulated Organizations**
  SSRN Preprint, 2026
  Explores how procedural and documentary infrastructures shape responsibility expectations prospectively, before action unfolds.
- STS **Structuring Judgment Under Constraint: Cognitive Friction in Procedural Texts**
  SSRN Preprint, 2026
  Examines how structural properties of procedural language shape interpretive burden and downstream reasoning demands.
- HCI **How Regulated Systems Produce Responsibility: A Four-Layer Framework for Interaction Design**
  SSRN Preprint, 2026
  Proposes a four-layer framework (accessibility, structure, coordination, reasoning) for analyzing regulated systems as interaction environments.
- HCI **When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation**
  SSRN Preprint, 2026
  Investigates loopback behavior as a structural signal of document system limitations rather than individual error.
- HCI **From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work**
  SSRN Preprint, 2026
  Examines how access constraints shape the set of actions practitioners consider during task execution.

- **HCI** **Friction Before Failure: Making Review Burden Visible in Regulated Document Work**
  SSRN Preprint, 2026
  Investigates how visibility of workflow friction alters coordination and responsibility perception in document-based review processes.

PROFESSIONAL EXPERIENCE

Quality Control / Quality Systems Project Lead Sep 2022 – Present
Pharmaceutical Manufacturing Company · South Korea

- Managed 20+ deviation and CAPA investigations end-to-end in GMP-regulated environments, observing how documentation structure shapes which causal explanations persist under institutional pressure.
- Supported 10+ regulatory inspections and global audits, developing practical insight into how traceability and accountability are operationalized through documentation systems.
- Revised and maintained 50+ SOPs across validation, deviation, and CAPA workflows, informing research on cognitive friction and interpretive variability in procedural texts.
- Designed documentation frameworks for pharmaceutical technology transfer, contributing to the structuring of accountability in cross-organizational processes.

METHODS & SKILLS

Qualitative	Think-aloud protocol; qualitative coding; document analysis; annotation protocol design; causal reasoning analysis
Quantitative	Experimental design; behavioral logging; NLP / text processing; machine learning (classification, validation); graph analysis; composite index construction
Tools	Python · R · Streamlit · LaTeX · HTML/CSS · GitHub
Languages	Korean (native) · English (fluent) · Japanese (fluent)